Block practice · Identities, equations and inequalities

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Linear inequalities and the 'both' overlap

Greatest or least value in a region

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Practice workbook · Linear inequalities and the 'both' overlap

Solving a single inequality in which x appears only to the first power, including one with surd coefficients.

15 question parts: 1 worked-example retry and 14 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-01

1. Question 1 · part a · 2 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part a only.

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1 Find the set of values of x for which

(a) $2(3x - 1) < 4 - 3x$

(2)

(b) $3x^2 - 8x - 3 < 0$

(4)

(c) **both** $2(3x - 1) < 4 - 3x$ **and** $3x^2 - 8x - 3 < 0$

(1)

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(Total for Question 1 is 7 marks)

1. Question 1 · part a · 2 marks · Working and answer

Working and answer

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2. Question 1 · part a · 1 mark

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

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1 Find the set of values for x for which

(a) $5x - 10 > 4x - 7$

(1)

(b) $2x^2 - 11x + 5 < 0$

(3)

(c) **both** $5x - 10 > 4x - 7$ **and** $2x^2 - 11x + 5 < 0$

(1)

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(Total for Question 1 is 5 marks)

2. Question 1 · part a · 1 mark · Working and answer

Working and answer

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3. Question 2 · part a · 1 mark

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

2 Find the set of values of x for which

(a) $3x - 8 < 5x + 3$

(1)

(b) $4x^2 - 7x + 1 > 6 - 2x^2$

(4)

(c) **both** $3x - 8 < 5x + 3$ **and** $4x^2 - 7x + 1 > 6 - 2x^2$

(1)

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3. Question 2 · part a · 1 mark · Working and answer

Working and answer

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4. Question 2 · part a · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part a only.

2 Find the set of values for x for which

(a) $9-3x>11x+2$

(1)

(b) $10x^2+7x<12$

(3)

(c) **both** $9-3x>11x+2$ **and** $10x^2+7x<12$

(1)

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4. Question 2 · part a · 1 mark · Working and answer

Working and answer

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5. Question 1 · part a · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

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1 Find the set of values for x for which

(a) $8x - 7 < 5x + 5$

(2)

(b) $2x^2 - 5x - 3 > 0$

(3)

(c) **both** $8x - 7 < 5x + 5$ **and** $2x^2 - 5x - 3 > 0$

(1)

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(Total for Question 1 is 6 marks)

5. Question 1 · part a · 2 marks · Working and answer

Working and answer

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6. Question 2 · part a · 2 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

2 Find the set of values of x for which

(a) $2(x+1) < 5x - 2$

(2)

(b) $3x^2 - x \leqslant 10$

(3)

(c) **both** $2(x+1) < 5x - 2$ **and** $3x^2 - x \leqslant 10$

(1)

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6. Question 2 · part a · 2 marks · Working and answer

Working and answer

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7. Question 2 · part a · 1 mark

2020 JAN · Paper 02

Further practice · Attempt part a only.

2 Find the set of values of x for which

(a) $3 + 2x \leqslant x + 2$

(1)

(b) $8x^2 + 10x < 3$

(4)

(c) **both** $3 + 2x \leqslant x + 2$ **and** $8x^2 + 10x < 3$

(1)

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P 5 9 9 3 9 A 0 4 3 6

7. Question 2 · part a · 1 mark · Working and answer

Working and answer

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8. Question 1 · part whole · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

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1 Without using a calculator, solve the inequality $\sqrt{50}\,x - \sqrt{18} > 6x + 5$

Give your answer in an exact form with a rationalised denominator.
Show your working clearly.

(4)

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(Total for Question 1 is 4 marks)

8. Question 1 · part whole · 4 marks · Working and answer

Working and answer

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9. Question 1 · part c · 1 mark

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

1 Find the set of values for x for which

(a) $5x - 10 > 4x - 7$

(1)

(b) $2x^2 - 11x + 5 < 0$

(3)

(c) **both** $5x - 10 > 4x - 7$ **and** $2x^2 - 11x + 5 < 0$

(1)

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(Total for Question 1 is 5 marks)

9. Question 1 · part c · 1 mark · Working and answer

Working and answer

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10. Question 2 · part c · 1 mark

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

2 Find the set of values of x for which

(a) $3x - 8 < 5x + 3$

(1)

(b) $4x^2 - 7x + 1 > 6 - 2x^2$

(4)

(c) **both** $3x - 8 < 5x + 3$ **and** $4x^2 - 7x + 1 > 6 - 2x^2$

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10. Question 2 · part c · 1 mark

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 2 continued

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(Total for Question 2 is 6 marks)

10. Question 2 · part c · 1 mark · Working and answer

Working and answer

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11. Question 1 · part c · 1 mark

2023 JAN · Paper 02

Further practice · Attempt part c only.

1 Find the set of values of x for which

(a) $2(3x - 1) < 4 - 3x$

(2)

(b) $3x^2 - 8x - 3 < 0$

(4)

(c) **both** $2(3x - 1) < 4 - 3x$ **and** $3x^2 - 8x - 3 < 0$

(1)

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(Total for Question 1 is 7 marks)

11. Question 1 · part c · 1 mark · Working and answer

Working and answer

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12. Question 2 · part c · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part c only.

2 Find the set of values for x for which

(a) $9-3x>11x+2$

(1)

(b) $10x^2+7x<12$

(3)

(c) **both** $9-3x>11x+2$ **and** $10x^2+7x<12$

(1)

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12. Question 2 · part c · 1 mark

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Further practice · Attempt part c only.

Question 2 continued

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(Total for Question 2 is 5 marks)

12. Question 2 · part c · 1 mark · Working and answer

Working and answer

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13. Question 1 · part c · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part c only.

1 Find the set of values for x for which

(a) $8x - 7 < 5x + 5$

(2)

(b) $2x^2 - 5x - 3 > 0$

(3)

(c) **both** $8x - 7 < 5x + 5$ **and** $2x^2 - 5x - 3 > 0$

(1)

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(Total for Question 1 is 6 marks)

13. Question 1 · part c · 1 mark · Working and answer

Working and answer

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14. Question 2 · part c · 1 mark

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

2 Find the set of values of x for which

(a) $2(x+1) < 5x - 2$

(2)

(b) $3x^2 - x \leqslant 10$

(3)

(c) **both** $2(x+1) < 5x - 2$ **and** $3x^2 - x \leqslant 10$

(1)

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14. Question 2 · part c · 1 mark

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 2 continued

(Total for Question 2 is 6 marks)

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14. Question 2 · part c · 1 mark · Working and answer

Working and answer

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15. Question 2 · part c · 1 mark

2020 JAN · Paper 02

Further practice · Attempt part c only.

2 Find the set of values of x for which

(a) $3 + 2x \leqslant x + 2$

(1)

(b) $8x^2 + 10x < 3$

(4)

(c) **both** $3 + 2x \leqslant x + 2$ **and** $8x^2 + 10x < 3$

(1)

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15. Question 2 · part c · 1 mark

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Further practice · Attempt part c only.



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15. Question 2 · part c · 1 mark · Working and answer

Working and answer

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Practice workbook · Greatest or least value in a region

Simple linear programming: finding the greatest or least value of an expression such as P = 2x + y over the region R.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-07

1. Question 5 · part c · 3 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part c only.

5 (a) On the grid opposite draw the line with equation

(i) $y = 2x + 5$ (ii) $4y = x - 8$ (iii) $5y + 3x = 30$

(3)

(b) Show, by shading, the region R defined by the inequalities

$y \leqslant 2x + 5$ $4y \geqslant x - 8$ $5y + 3x \leqslant 30$

(1)

For all points in R with coordinates (x, y)

$P = 2x - 5y$

(c) Using your graph, find the least value of P

(3)

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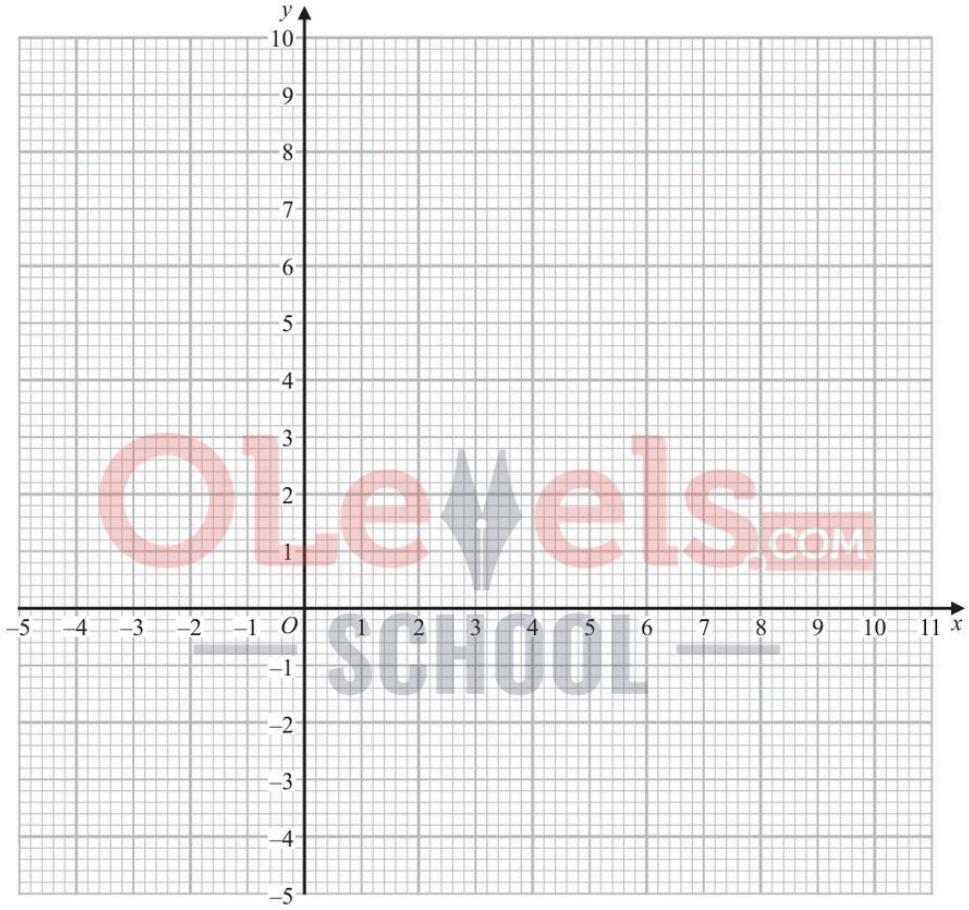
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1. Question 5 · part c · 3 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part c only.

Question 5 continued



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Turn over for a spare grid if you need to redraw your graph.

1. Question 5 · part c · 3 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part c only.

Question 5 continued

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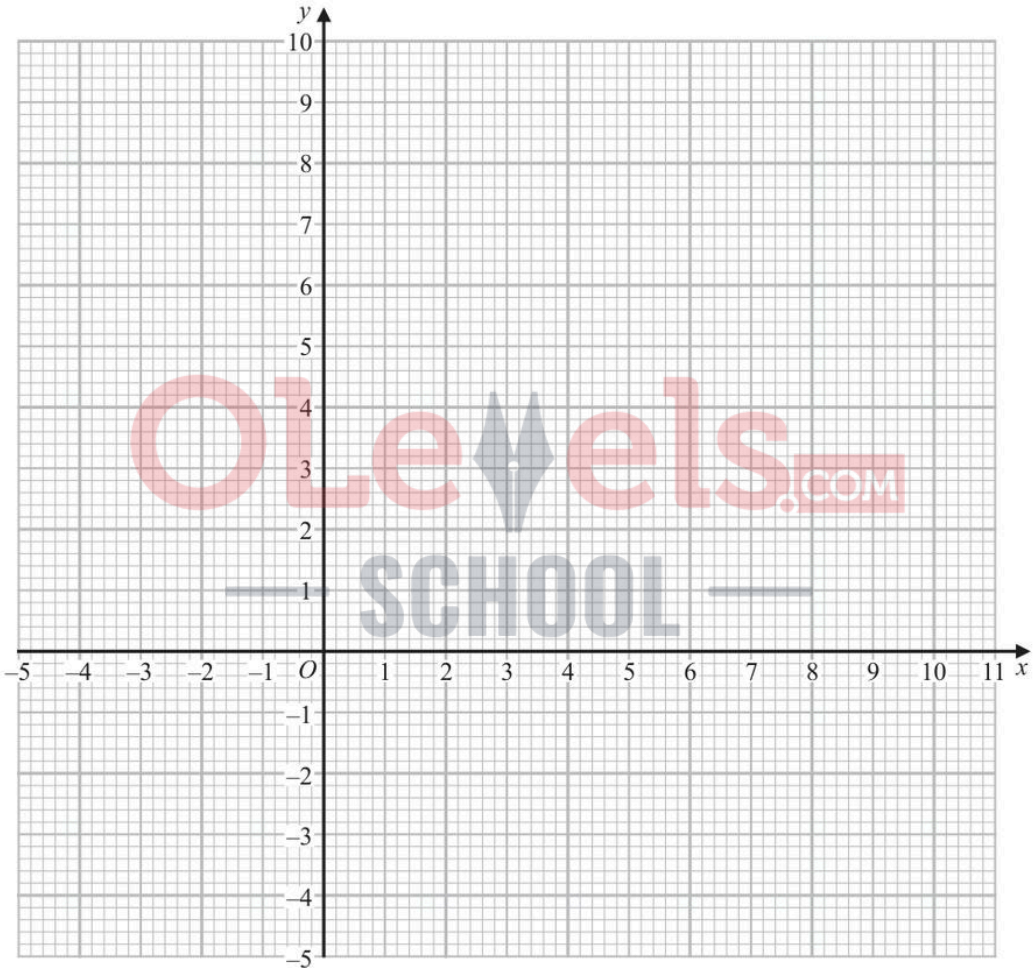
1. Question 5 · part c · 3 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part c only.

Question 5 continued

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(Total for Question 5 is 7 marks)

1. Question 5 · part c · 3 marks · Working and answer

Working and answer

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2. Question 4 · part c · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part c only.

4 (a) On the grid opposite, draw the line with equation

(i) $y = 2x - 4$ (ii) $2x + 3y = 12$ (iii) $y + 2x + 2 = 0$

(3)

(b) Show, by shading on the grid, the region R defined by the inequalities

$y \geqslant 2x - 4$ $2x + 3y \leqslant 12$ $y + 2x + 2 \geqslant 0$

(1)

For all points in R, with coordinates (x, y)

$$P = x - 2y$$

(c) find the least value of P

(4)

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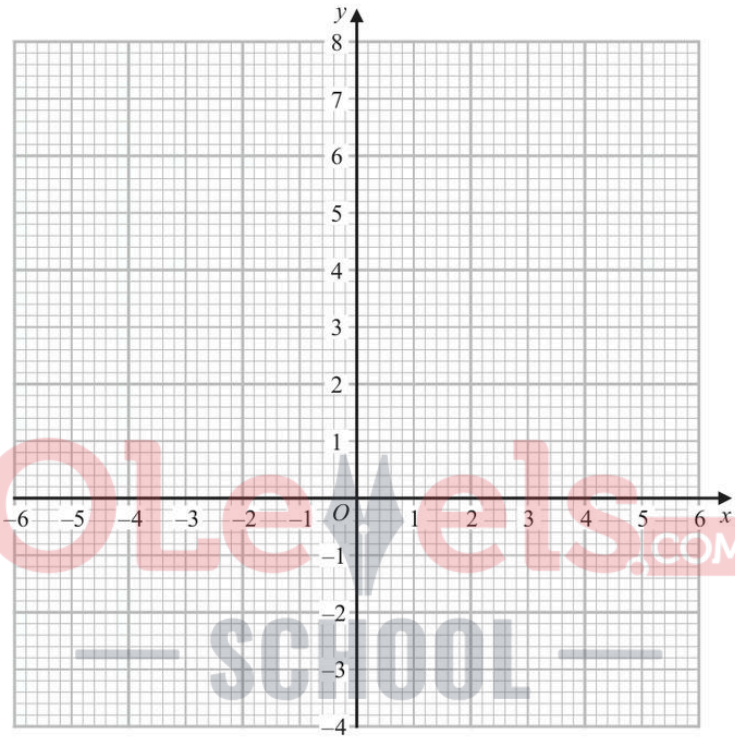
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2. Question 4 · part c · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part c only.

Question 4 continued



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P 6 6 3 1 0 A 0 9 3 6

9

Turn over ▶

2. Question 4 · part c · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part c only.

Question 4 continued

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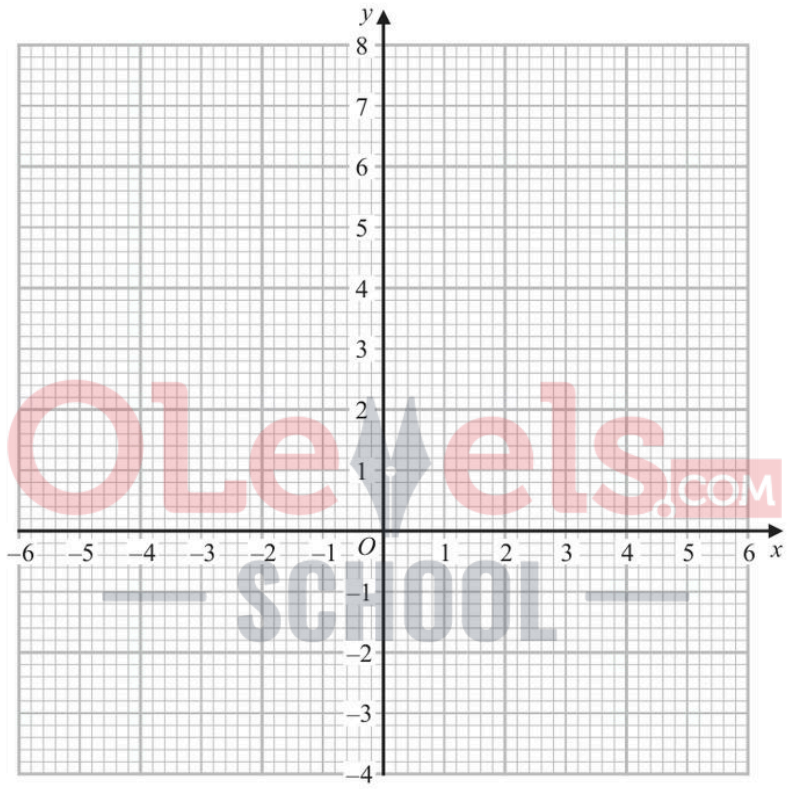
2. Question 4 · part c · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part c only.

Question 4 continued

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(Total for Question 4 is 8 marks)

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2. Question 4 · part c · 4 marks · Working and answer

Working and answer

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3. Question 4 · part c.ii · 4 marks

2023 NOV · Paper 02

Further practice · Attempt part c.ii only.

4 (a) On the axes opposite, draw the line with equation

(i) $y = -x - 1$ (ii) $y - 3x + 8 = 0$ (iii) $2y = x + 8$

(3)

(b) Show, by shading on your graph, the region R defined by the inequalities

$y \geqslant -x - 1$ and $y \geqslant 3x - 8$ and $2y \leqslant x + 8$

(1)

For all points in R, with coordinates (x, y)

$$P = 2y - 3x$$

(c) Find

(i) the greatest value of P

(ii) the least value of P

(4)

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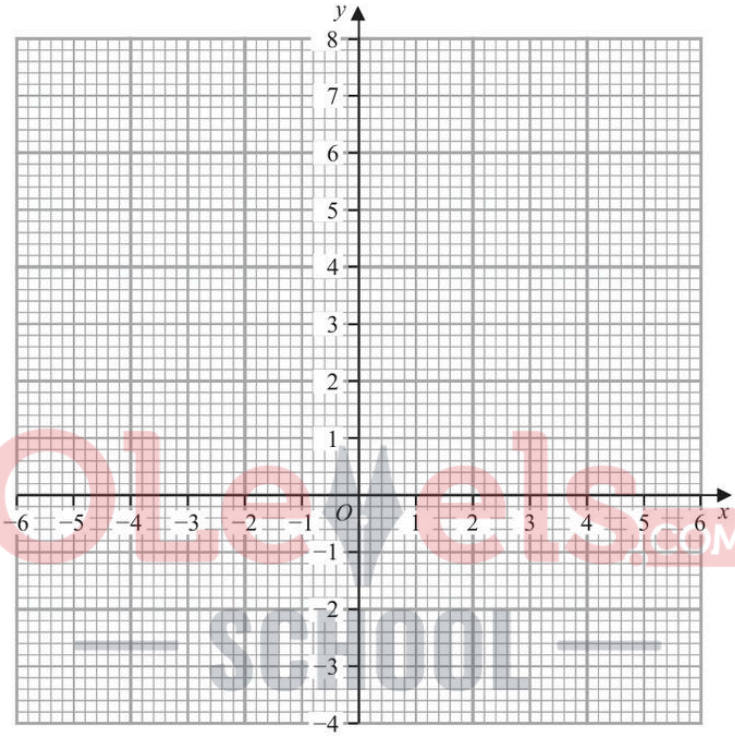
P 7 3 5 8 6 A 0 8 3 2

3. Question 4 · part c.ii · 4 marks

2023 NOV · Paper 02

Further practice · Attempt part c.ii only.

Question 4 continued



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Turn over for a spare grid if you need to redraw your graph.

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Turn over

3. Question 4 · part c.ii · 4 marks

2023 NOV · Paper 02

Further practice · Attempt part c.ii only.

Question 4 continued

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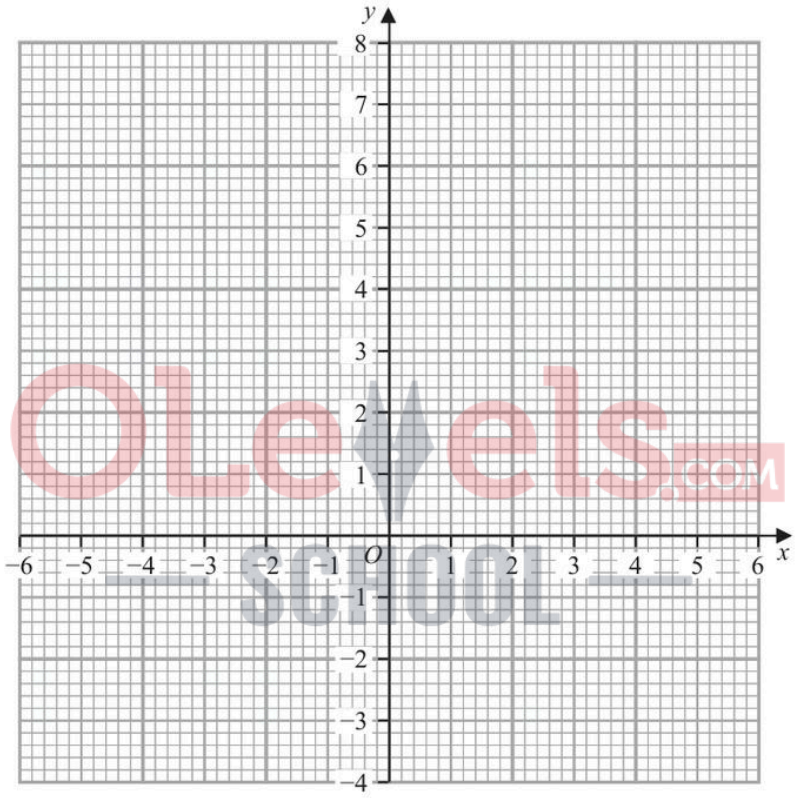
3. Question 4 · part c.ii · 4 marks

2023 NOV · Paper 02

Further practice · Attempt part c.ii only.

Question 4 continued

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(Total for Question 4 is 8 marks)

3. Question 4 · part c.ii · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 4 · part c · 4 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

4 (a) On the grid opposite, draw the graph of the line with equation

(i) $3x+2y=18$ (ii) $x+3y-6=0$ (iii) $y=3x$

(3)

(b) Show, by shading on the grid, the region R defined by the inequalities

$3x+2y \leqslant 18$ $x+3y-6 \geqslant 0$ $y \leqslant 3x$

(1)

For all points in R, with coordinates (x, y)

$$P=2x-y$$

(c) find the least value of P

(4)

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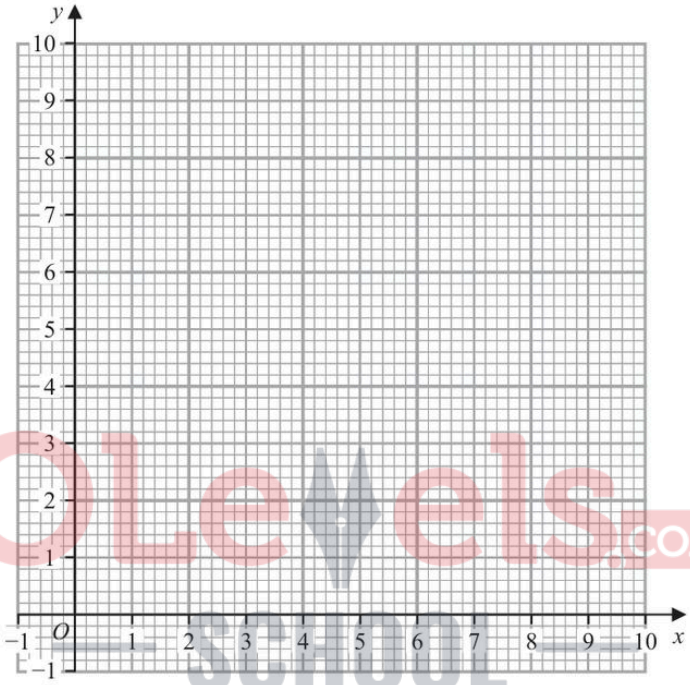
P 7 2 8 6 5 A 0 1 2 3 6

4. Question 4 · part c · 4 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

Question 4 continued



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Turn over for a spare grid if you need to redraw your graph.

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4. Question 4 · part c · 4 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

Question 4 continued

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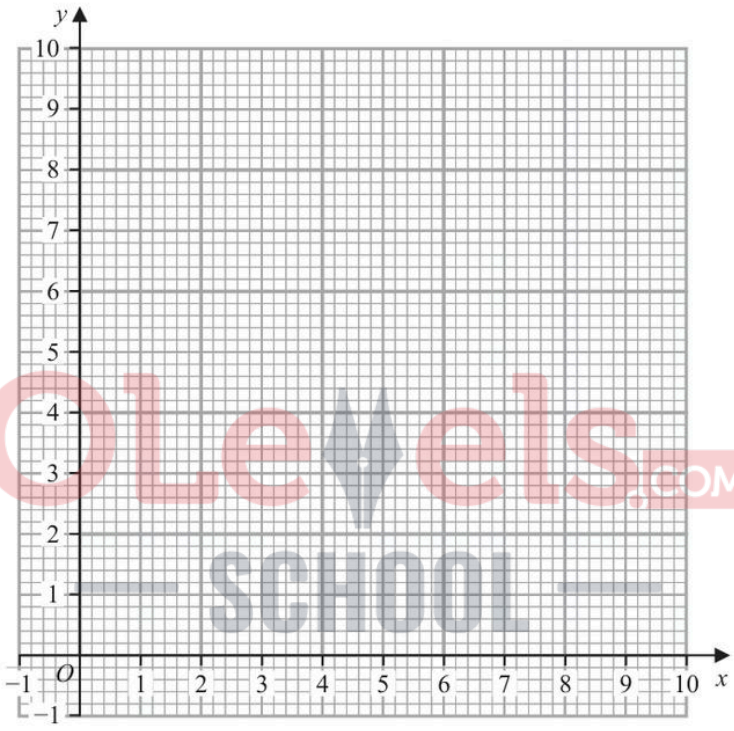
4. Question 4 · part c · 4 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

Question 4 continued

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(Total for Question 4 is 8 marks)

4. Question 4 · part c · 4 marks · Working and answer

Working and answer

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5. Question 5 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

5 (a) On the grid opposite, draw the graphs of the lines with equations

$$2x + 3y = 24 \qquad y = 2x \qquad 3y = 2x - 12$$

(3)

(b) Show, by shading on the grid, the region R defined by the inequalities

$$2x + 3y \leqslant 24 \qquad y \leqslant 2x \qquad 3y \geqslant 2x - 12 \qquad y \geqslant 0$$

(1)

For all points in R, with coordinates (x, y)

$$F = 2x + 5y$$

(c) Find the greatest value of F.

(3)

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5. Question 5 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

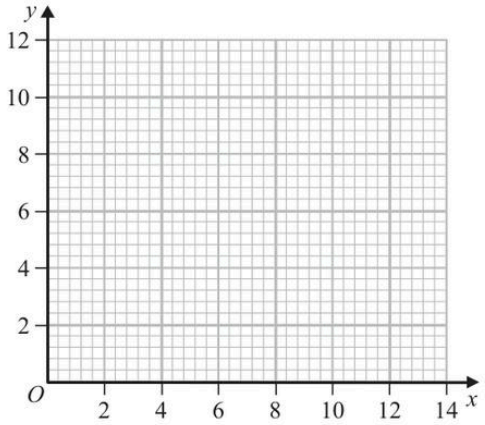
Further practice · Attempt part c only.

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Question 5 continued



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Turn over for a spare grid if you need to redraw your graphs.

5. Question 5 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 5 continued

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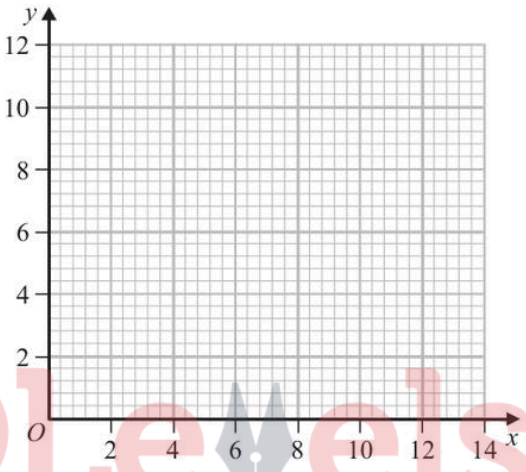
5. Question 5 · part c · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 5 continued

Only use this grid if you need to redraw your graphs.



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(Total for Question 5 is 7 marks)

5. Question 5 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 5 · part c · 4 marks

2025 NOV · Paper 01

Further practice · Attempt part c only.

5 (a) On the grid opposite, draw the line with equation

(i) $3x-4y=12$ (ii) $y+6+3x=0$ (iii) $3y=18-x$

(3)

(b) Show, by shading on the grid, the region R defined by the inequalities

$3x-4y \leqslant 12$ $y+6+3x \geqslant 0$ $3y \leqslant 18-x$

(1)

For all points in R, with coordinates (x, y)

$$P=3x-2y$$

Using values from your graph,

(c) find the least value of P and the greatest value of P

(4)

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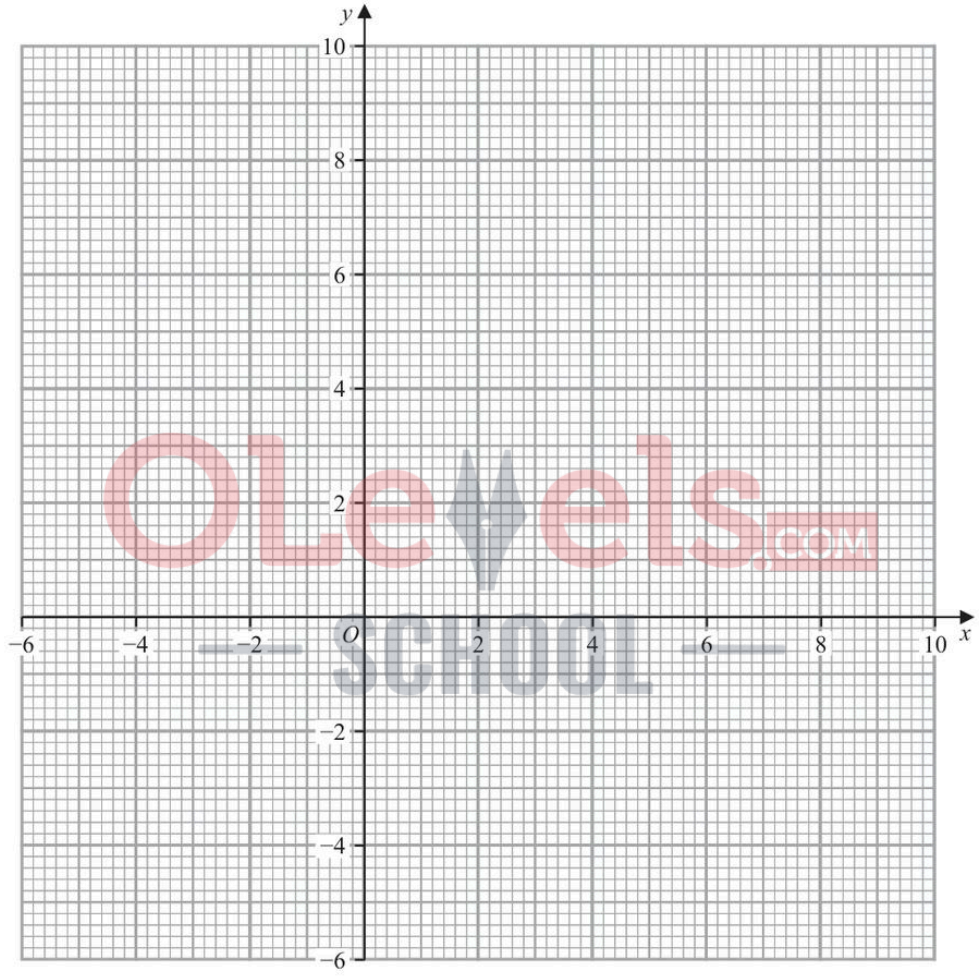
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6. Question 5 · part c · 4 marks

2025 NOV · Paper 01

Further practice · Attempt part c only.

Question 5 continued



DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

Turn over for a spare grid if you need to redraw your graph.

6. Question 5 · part c · 4 marks

2025 NOV · Paper 01

Further practice · Attempt part c only.

Question 5 continued

DO NOT WRITE IN THIS AREA

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6. Question 5 · part c · 4 marks

2025 NOV · Paper 01

Further practice · Attempt part c only.

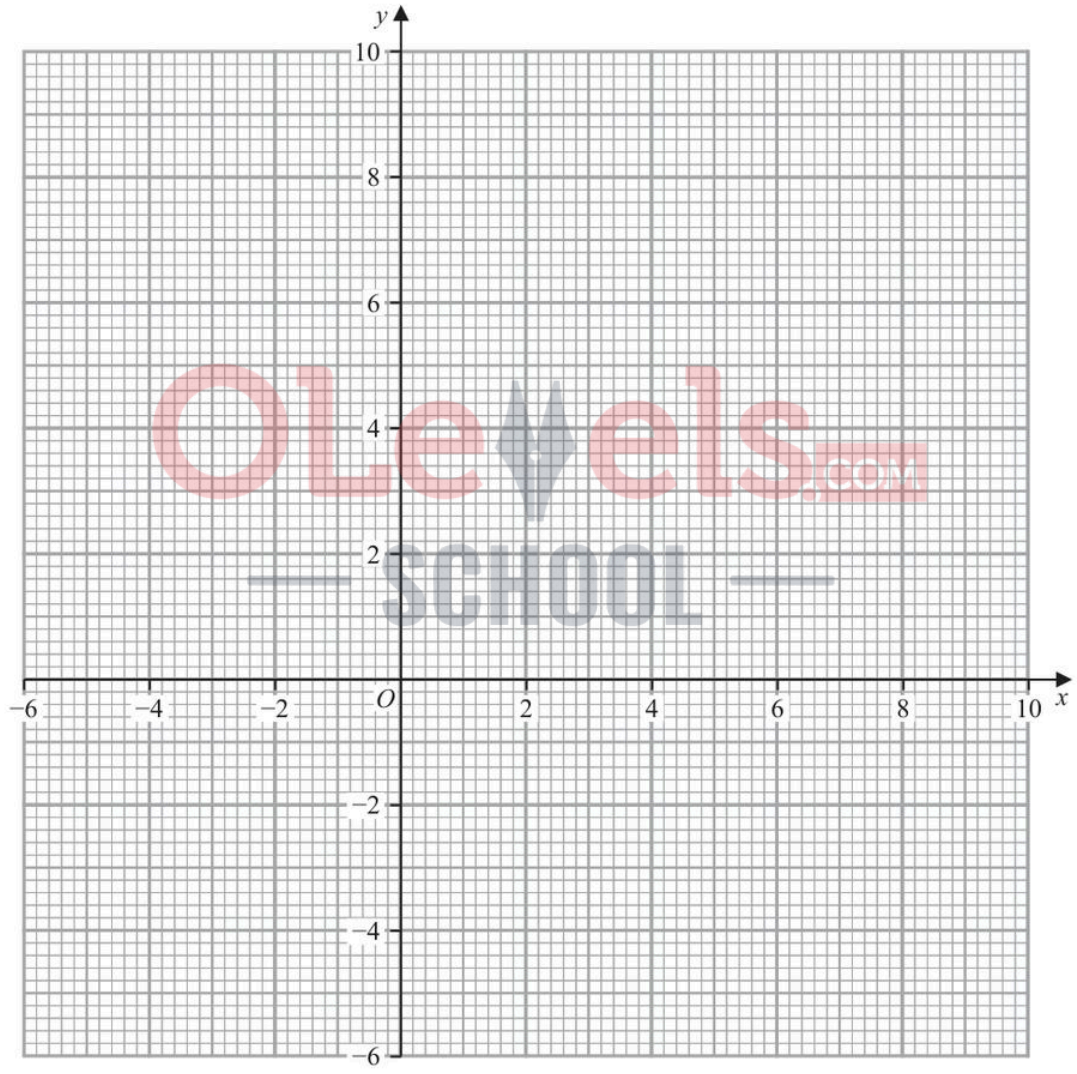
DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

Question 5 continued

Only use this grid if you need to redraw your graph.



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(Total for Question 5 is 8 marks)

6. Question 5 · part c · 4 marks · Working and answer

Working and answer

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